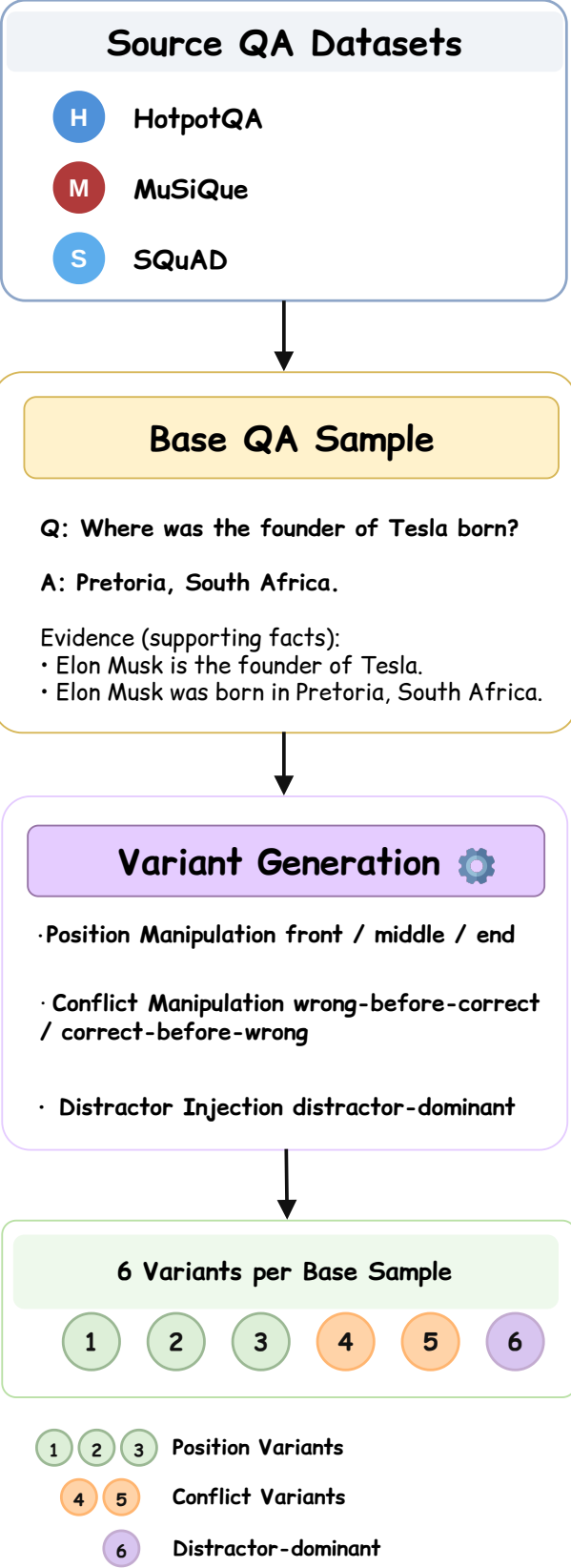


1) Benchmark Construction



Base QA Sample

Q: Where was the founder of Tesla born?

A: Pretoria, South Africa.

Evidence (supporting facts):

•

Elon Musk is the founder of Tesla.

•

Elon Musk was born in Pretoria, South Africa.

Variant Generation

•

Position Manipulation front / middle / end

•

Conflict Manipulation wrong-before-correct / correct-before-wrong

•

Distractor Injection distractor-dominant

6 Variants per Base Sample

1

2

3

4

5

6

1

2

3

Position Variants

4

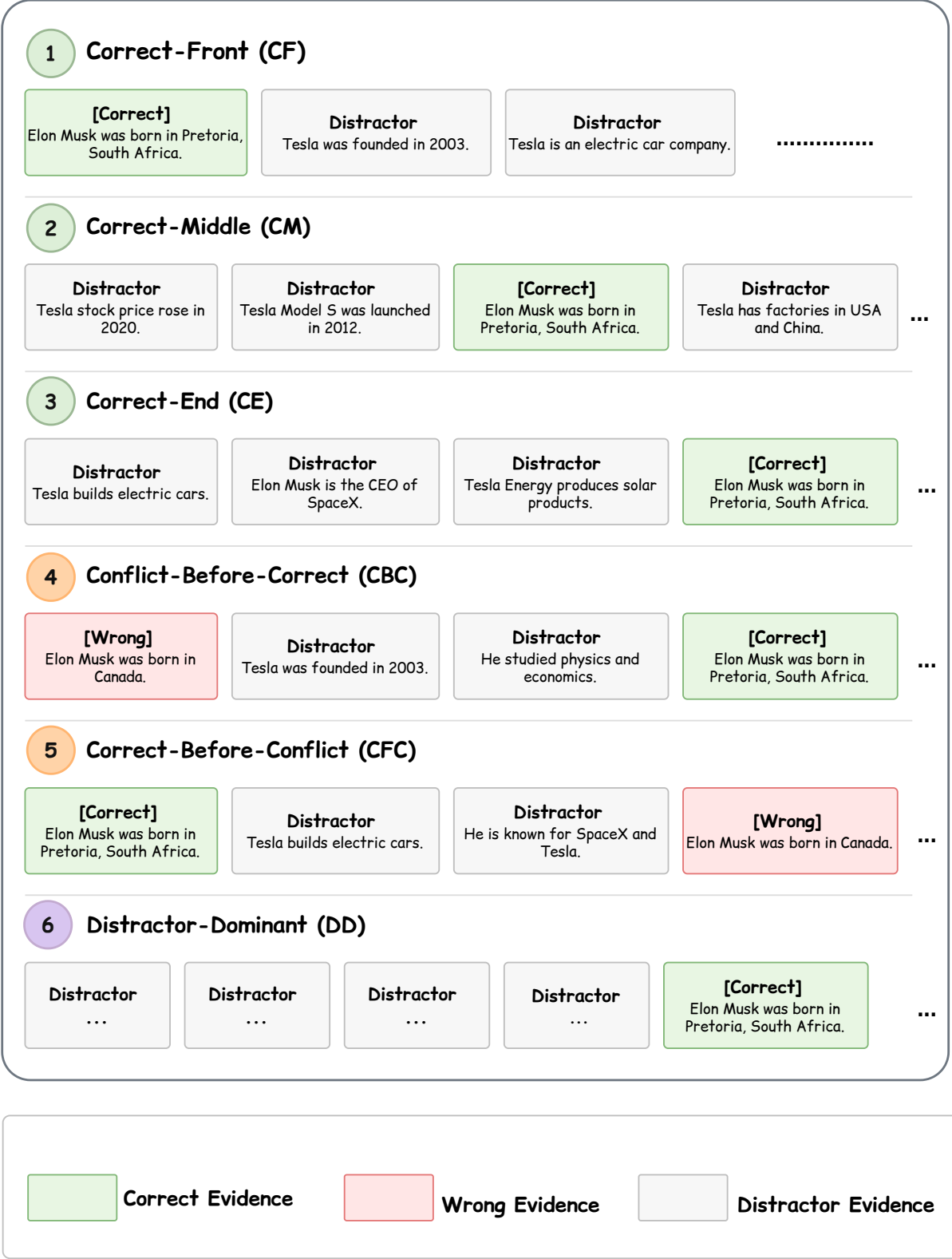
5

Conflict Variants

6

Distractor-dominant

2) Example: 6 Evidence-Layout Variants



Distractor

.....

2 Correct-Middle (CM)

Distractor

Distractor

[Correct]

Distractor

...

3 Correct-End (CE)

Distractor

Distractor

Distractor

[Correct]

...

4 Conflict-Before-Correct (CBC)

[Wrong]

Distractor

Distractor

[Correct]

...

5 Correct-Before-Conflict (CFC)

[Correct]

Distractor

Distractor

[Wrong]

...

6 Distractor-Dominant (DD)

Distractor

Distractor

Distractor

Distractor

[Correct]

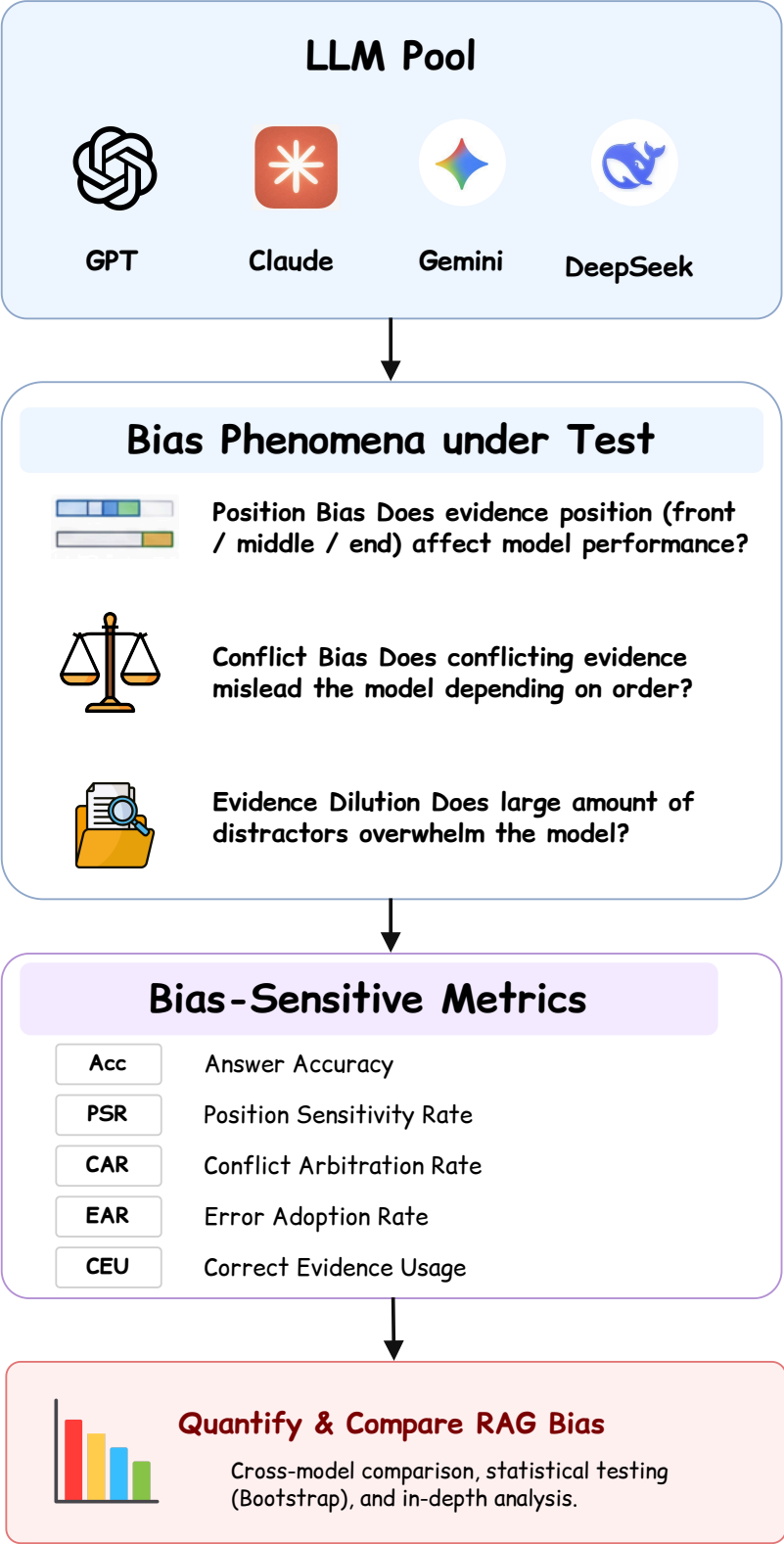
...

Correct Evidence

Wrong Evidence

Distractor Evidence

3) Multi-LLM Evaluation



Bias-Sensitive Metrics

Acc

Answer Accuracy

PSR

Position Sensitivity Rate

CAR

Conflict Arbitration Rate

EAR

Error Adoption Rate

CEU

Correct Evidence Usage

Quantify & Compare RAG Bias

Cross-model comparison, statistical testing (Bootstrap), and in-depth analysis.